YAKEEN

NEET 2.0 2027

Physics | Lecture - 08

Units and Measurement

By- Manish Raj (MR Sir)



H/W Todays Goal

→ Vector → Maha manthan
Soln upladed ✓

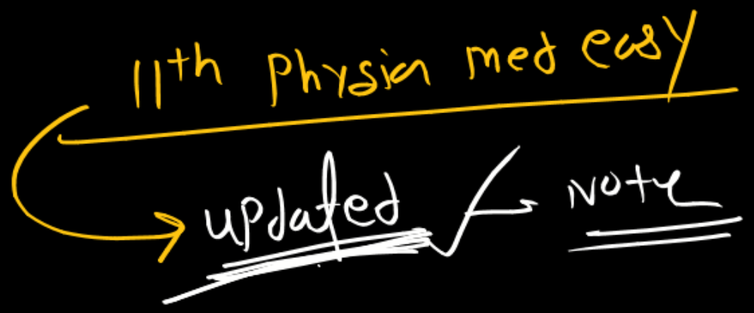
→ U.d.M → Maha Manthan ✓
Question → 108

→ DPP ✓

Todays Goal

JEE- Adv Questions ✓

Instrument


11th physics med easy
updated
note

$$\Rightarrow \frac{d \log_e x}{dx} = \frac{1}{x} \checkmark$$

$$\frac{d \log_e 2}{dx} = 0$$

$$\Rightarrow \frac{d \log_e (\sin x)}{dx} = \frac{1}{\sin x} \times \cos x = \cot x \checkmark$$

(inside)

$$\frac{d \log_e (1-a)}{da} = \frac{1}{(1-a)} \times (-1)$$

$$= \frac{-1}{1-a} \checkmark$$

$$\Rightarrow \frac{d \log_e (1+x^2)}{dx} = \frac{1}{1+x^2} \times (0+2x) = \frac{2x}{1+x^2}.$$

$$\Rightarrow \frac{d \log_e (1+x^2)}{dt} = \frac{2x}{1+x^2} \frac{dx}{dt} \checkmark$$

Question

Using the expression $2d \sin\theta = \lambda$, one calculates the values of d by measuring the corresponding angles θ in the range 0 to 90°. The wavelength λ is exactly known and the error in θ is constant for all values of θ. As increases from 0°:

[JEE-Advance 2013]

1. the absolute error in d remains constant
2. the absolute error in d increases
3. the fractional error in d remains constant
4. ✓ the fractional error in d decreases

$$d = \frac{\lambda}{2\sin\theta}$$ taking log both side

$$\log d = \log\left(\frac{\lambda}{2\sin\theta}\right)$$

$$\log d = \log\lambda - \log 2 - \log(\sin\theta)$$

diff^n w.r.t 't'

$$\frac{1}{d}\frac{\partial d}{\partial t} = 0 - 0 - \frac{1}{\sin\theta}\cos\theta\frac{d\theta}{dt}$$

$$\left|\frac{\partial d}{d}\right| = +\cot\theta\, d\theta \rightarrow \theta = 0° \text{ to } 90°$$

$$\frac{\partial d}{dt} = 0\,(2\sin\theta) - \frac{2}{2}\cos\theta\frac{d\theta}{dt}$$

Question

Consider the ratio $r = \frac{(1-a)}{(1+a)}$ to be determined by measuring a dimensionless quantity a. If the error in the measurement of a is Δa ($\Delta a/a << 1$), then what is the error Δr is determining r? **(JEE Adv. 2018)**

1. $\frac{\Delta a}{(1+a)^2}$
2. $\frac{+2\Delta a}{(1+a)^2}$
3. $\frac{2\Delta a}{(1-a)^2}$
4. $\frac{2a\Delta a}{(1-a^2)}$

method-1

Δr = absolute error in r

$r = \frac{1-a}{1+a}$ ⇒ division

diff^n w.r.t 'a'

$\frac{dr}{da} = \frac{(-1)(1+a) - (1-a)[1]}{(1+a)^2}$

$dr = \frac{-1-a-1+a}{(1+a)^2}\,da$

$dr = \frac{+2\,da}{(1+a)^2}$ ← Ans 2

method-2

$r = \frac{1-a}{1+a} + 1 - 1$ (one add & one subtr)

$r = \frac{1-a+1+a}{1+a} - 1$

$r = \frac{2}{1+a} - 1$

diff^n w.r.t 'a'

$\frac{dr}{da} = \frac{-2(0+1)}{(1+a)^2} - 0$

$dr = \frac{-2\,da}{(1+a)^2}$ ✓

copy at ghz

method - 3

$$\gamma = \frac{1-a}{1+a}$$

taking log both side

$$\log \gamma = \log\left(\frac{1-a}{1+a}\right)$$

$$\log \gamma = \log(1-a) - \log(1+a)$$

diff^n w.r.t a

$$\frac{1}{\gamma}\frac{d\gamma}{da} = \frac{1}{1-a}(-1) - \frac{1}{(1+a)}(1)$$

$$\frac{1}{\gamma}\frac{d\gamma}{da} = \frac{-1}{1-a} - \frac{1}{1+a}$$

$$\frac{d\gamma}{da} = \gamma\left[\frac{-1-a-1+a}{(1-a)(1+a)}\right]$$

Put (γ)

$$= \frac{1-a}{1+a}\left[\frac{-2}{(1-a)(1+a)}\right]$$

$$d\gamma = \frac{-2\,da}{(1+a)^2}$$

$$\gamma = \frac{1-a}{1+a}$$

taking log both side

Question

The energy of a system as a function of time t is given as $E(t) = A^2 \exp(-\alpha t)$, where $\alpha = 0.2 s^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%, the percentage error in the value of $E(t)$ at $t = 5s$ is

(JEE Adv. 2015)

$E = A^2 e^{-\alpha t}$

taking log both side

$\log E = \log (A^2 e^{-\alpha t})$

$\log_e E = \log_e (A)^2 + \log_e e^{-\alpha t}$

$\log_e E = 2 \log_e (A) - \alpha t \log_e e^{1}$

$\log_e E = 2 \log_e A - \alpha t$

$E = A^2 e^{-\alpha t}$

Postulate

$\frac{\Delta A}{A} \times 100 = 1.25\%$

$2 \times \frac{\Delta t}{T} = 1.5\%$

diff^n w.r.t. time

$\frac{1}{E}\frac{dE}{dt} = 2\frac{1}{A}\frac{dA}{dt} - \alpha \quad (1)$

$100 \times \frac{\delta E}{E} = 2\left[\frac{dA}{A} \times 100\right] - \alpha\, dt \times 100$

$= 2 \times 1.25 + \frac{2}{10} 1.5 \times T$

$= 2.5 + \frac{3}{10} \times 5 = 2.5 + 1.5$

$= 4\%$

JEE Advance

phy

$$E = A^2 e^{-\alpha t}$$

Apply Product Rule of diff^n w.r.t. t

$$\frac{dE}{dt} = \left(\frac{dA^2}{dt}\right) e^{-\alpha t} + A^2 \frac{de^{-\alpha t}}{dt}$$

$$\frac{dE}{dt} = 2A \frac{dA}{dt} e^{-\alpha t} + A^2 e^{-\alpha t} (-\alpha)$$

$$dE = 2A e^{-\alpha t} dA + A^2 e^{-\alpha t} \alpha \, dt$$

divided by E both side

$$\left(\frac{dE}{E}\right) = \frac{2A e^{\alpha t} dA}{A^2 e^{-\alpha t}} + \frac{A^2 e^{-\alpha t} \alpha \, dt}{A^2 e^{-\alpha t}}$$

$$\boxed{\frac{dE}{E} = 2\left(\frac{dA}{A}\right) + \alpha \, dt} \quad *$$

Question

Copy

The current voltage relation of diode is given by $I = (e^{1000\,V/T} - 1)$ mA, where the applied voltage V is in volt and the temperature T is in kelvin. If a student makes an error measuring ±0.01 V while measuring the current of 5 mA at 300 K, what will be the error in the value of current in mA?

(JEE Adv. 2018)

1. 0.2 mA
2. 0.02 mA
3. 0.5 mA
4. 0.05 mA

$\Delta I = ??$

$\Delta V = 0.01$ volt

$I = 5$ mA

$T = 300$ K

Put 'I'

$5\text{mA} = e^{\frac{1000V}{T}} - 1$

$6\text{mA} = e^{\frac{1000V}{T}}$

$I = e^{\frac{1000V}{T}} - 1$

diff^n w.r.t 'V'

$\frac{dI}{dV} = e^{\frac{1000V}{T}} \times \frac{1000}{T} - 0$

$dI = e^{\frac{1000V}{T}} \times \frac{1000}{T}\, dV$

$= 6 \times \frac{300}{5} \times \frac{1}{1000} = \frac{1}{5} = 0.2\text{mA}$

Question

Try

In an experiment to determine the acceleration due to gravity g the formula used for the time period of a periodic motion is $T = 2\pi\sqrt{\frac{7(R-r)}{5g}}$. The values of R and r are measured to be (60 ± 1) mm and (10 ± 1) mm, respectively. In five successive measurements, the time period is found to be 0.52 s, 0.56 s, 0.57s, 0.54 s and 0.59 s. The least count of the watch used for the measurement of time period is 0.01 s. Which of the following statement(s) is (are) true? **[JEE-Advance 2016]**

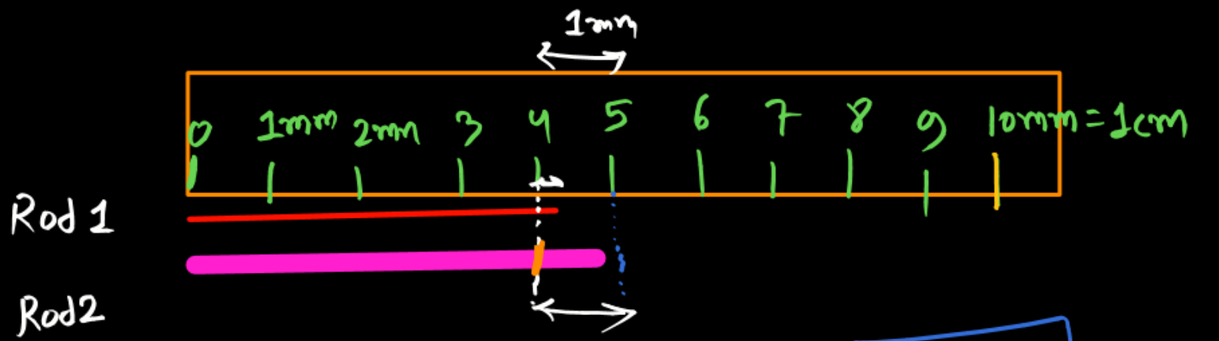
1. The error in the measurement of r is 10%
2. The error in the measurement of T is 3.57%
3. The error in the measurement of T is 2%
4. The error in the determined value of g is 11%

Measuring instrument :-

NEET

1. Metre scale
2. Vernier Callipers
3. Screw gauge (spherometre)

Metre scale



- Maximum error can be Produce by metre scale = 1mm
- Minimum reading taken by metre scale = 1mm

⇑

❤ least Count = L.C = 1mm ❤

length$_1$ = 4.2 mm (✗) 4.3mm (✗) Can't say (✗)

length$_2$ = 5mm (✗) 4.9mm (✗)

✓ 4 mm ❤ Answer

✓ 4mm ❤

final reading = (M.S.R ± L.C)
= (4mm ± 1mm)

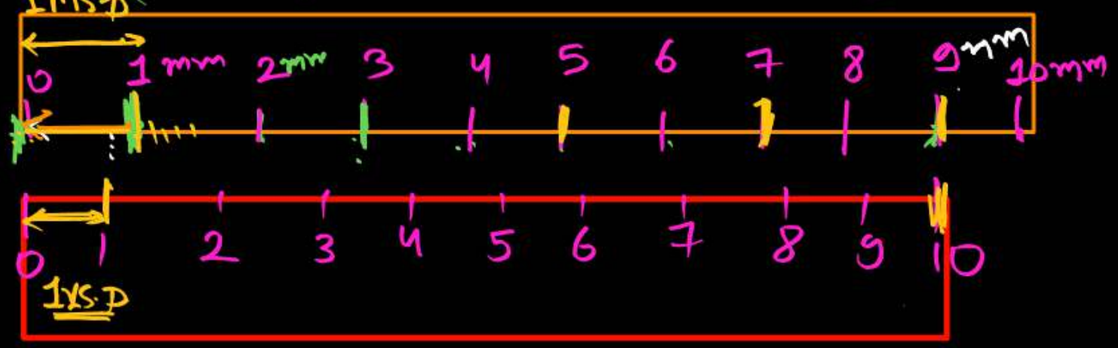
2. Vernier Callipers

L.C = minimum measurment

L.C = 1 M.S.D − 1 V.S.D

1 M.S.D = 1mm ✓

(fixed) main scale [metre scale]



vernier scale (movable)

10 V.S.D = 9 M.S.D = 9mm

vernier scale divisi.

$1\ V.S.D = \frac{9mm}{10}$

1 V.S.D = 0.9 mm

Question framing Boy :—

① Find L.C (M.S.D & V.S.D) gi.

1 M.S.D gi.

② Ready

Measuring Instrument

1. Metre Scale
2. Vernier Calliper
3. Screw gauge

Question

Main scale division is 1 cm and 9 MSD coinside with 12VSD then find LC?

$1\,MSD = 1\,cm$

$9\,M.S.D = 12\,V.S.D$

$L.C = 1\,MS.D - 1\,V.S.D$

$= 1\,M.S.D - \frac{9}{12}\,M.S.D$

$= \left(1 - \frac{9}{12}\right) MS.D$

$= \frac{3}{12} \times 1\,cm = \frac{1}{4}\,cm = 0.25\,cm$

Question

In certain vernier callipers, 25 divisions on the vernier scale have the same length as 24 divisions on the main scale. One division on the main scale is 1 mm long. The least count of the instrument is

1. 0.04 mm
2. 0.01 mm
3. 0.02 mm
4. 0.08 mm

1 m.SD = 1 mm

25 V.S.D = 24 m.S.D

1 VSD = $\frac{24}{25}$ m.SD

L.C = 1 MSD − 1 V.S.D

= 1 m.S.D − $\frac{24}{25}$ m.S.D

= $\frac{1}{25}$ MSD

= $\frac{1\,mm}{25}$

Question

In a Vernier calliper, one main scale division is x cm and n division of Vernier scale coincide with $(n-1)$ division of the main scale. The least count of the Vernier caliper in cm is:

1 m.s.D = x cm

n V.S.D = (n-1) M.S.D

L.C = ??

1. $\left(\frac{n-1}{n}\right)x$
2. $\frac{nx}{(n-1)}$
3. $\frac{x}{n}$
4. $\frac{x}{n-1}$

L.C = 1 M.S.D $-\left(\frac{n-1}{n}\right)$ M.S.D

$= \frac{n-n+1}{n}$ m

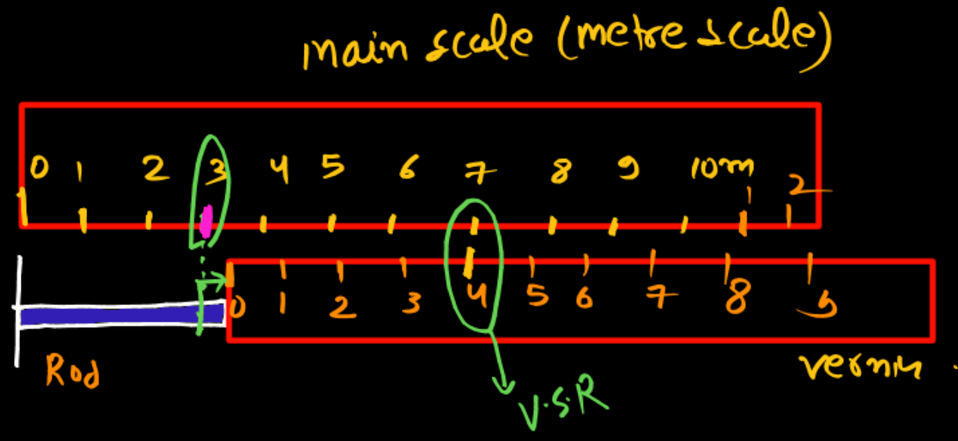
$= \frac{1}{n}x = \frac{x}{n}$ cm.

Question

No cap!

One cm on the main scale of vernier callipers is divided into ten equal parts. If 20 divisions of vernier scale coincide with 8 small divisions of the main scale. What will be the least count of callipers?

How To take reading.



$$\text{length of Rod} = \left[\text{M.S.R} + \text{V.S.R}(\text{L.C})\right]$$

(main scale Reading)

$$\text{final Ansl} = (3\text{mm} + 4 \times 0.1\text{mm})$$

$$= \{3.4\text{mm}\}$$

$$(l \pm \underbrace{\Delta l}_{\text{L.C}})$$

$$= (3.4\text{mm} \pm 0.1\text{mm})$$

Zero error

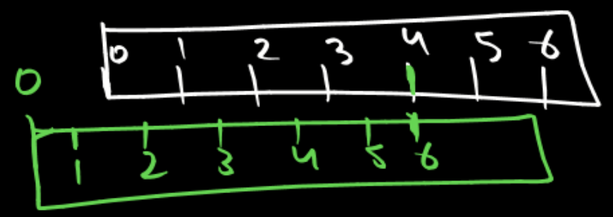
→ Zero error always subtracted with proper sign.

→ When we are not taking any reading ! —
but zero of main scale & vernier scale is not coinciding

-ve zero error

+ve zero error → zero of vernier scale right side of zero of main scale

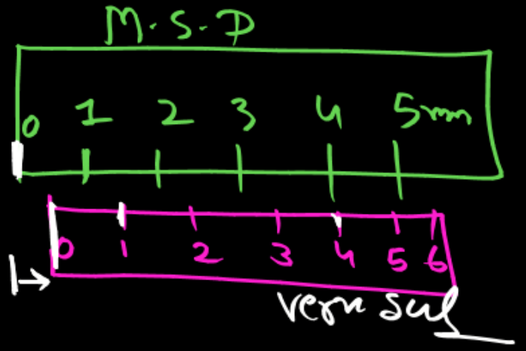
When zero of vernier scale left side of zero of main scale



Zero error = (10 - V.S.R) L.C

= - (10 - 6) × 0.1

= - 0.4 mm



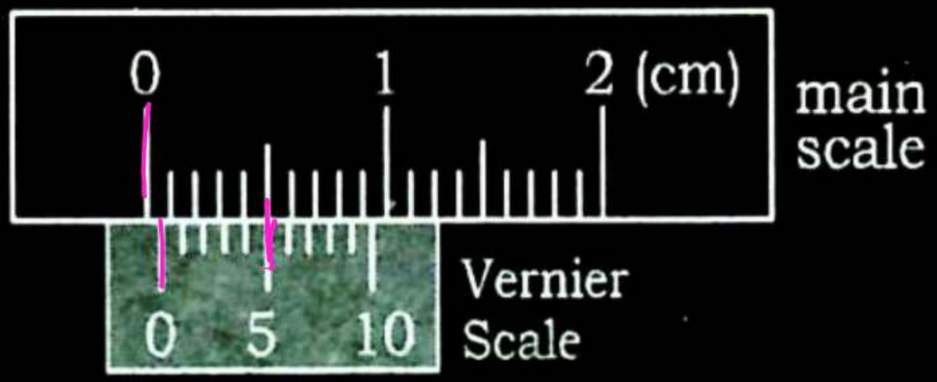
Zero error = +ve = + V.S.R (L.C)

= 0.4 mm


ZERO ERROR

Question

Find the zero correction in the given figure.



1. 0.4 mm
2. +0.5 mm
3. −0.5 mm
4. −0.4 mm

Question

A screw gauge has least count of 0.01 mm and there arc 50 divisions in its circular scale. The pitch of the screw gauge is [NEET-2021]

1. 0.01 mm
2. 0.25 mm
3. 0.5 mm
4. 1.0 mm

Question

The vernier scale used for measurement has a positive zero error of 0.2 mm. If while taking a measurement it was noted that zero on the vernier scale lies between 8.5 cm and 8.6 cm. and vernier coincidence is 6, then the correct value of measurement is _____ (least count = 0.01 cm) **[17 March, 2021 (Shift-I)]**

1. 8.58 cm
2. 8.54 cm
3. 8.56 cm
4. 8.36 cm

Question

10 mm = 1 cm
1 mm = 0.1 cm

0.1 cm = 1 mm

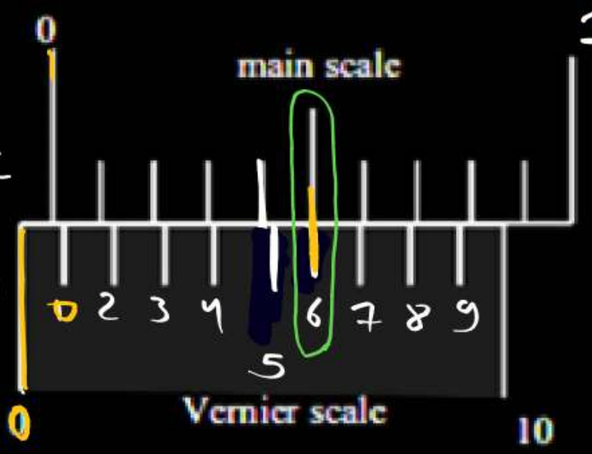
The smallest division on the main scale of a Vernier calipers is 0.1 cm. Ten divisions of the Vernier scale correspond to nine divisions of the main scale. The figure below on the left shows the reading of this caliper with no gap between its two jaws. The figure on the right shows the reading with a solid sphere held between the jaws. The correct diameter of the sphere is

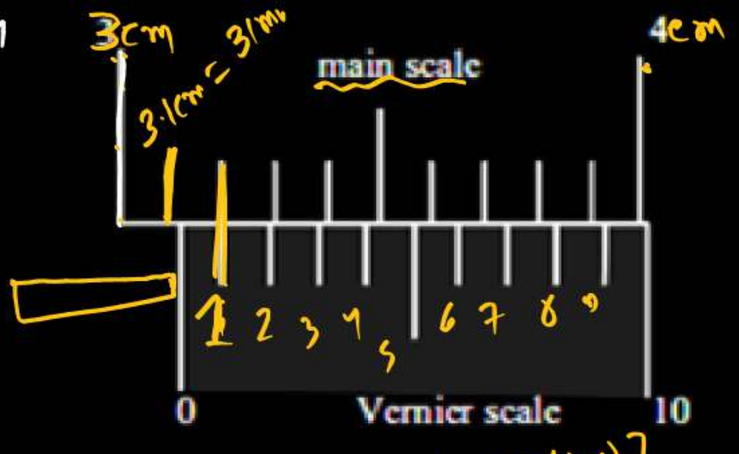
(JEE Adv. 2021)

1. 3.07 cm
2. 3.11 cm
3. 3.15 cm
4. 3.17 cm

L.C = 0.1 mm = 0.01 cm

Zero error = (10 − 6) × L.C
= 4 × 0.01 cm
= − 0.04 cm





final reading = 3.11 cm − (− 0.04 cm)
= 3.15 cm

Reading = [M.S.R + V.S.R (L.C)]
= 3.1 cm + 1 × 0.01 cm
= 3.11 cm

Question

The diameter of a cylinder is measured using a vernier calipers with no zero error. It is found that the zero of the vernier scale lies between 5.10 cm and 5.15 cm of the main scale. The vernier scale has 50 division equivalent to 2.45 cm. The 24th division of the vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is **(JEE Adv. 2013)**

1. 5.112 cm
2. 5.124 cm
3. 5.136 cm
4. 5.148 cm

Question H/W

A vernier calipers has 1 mm marks on the main scale. It has 20 equal divisions on the vernier scale which match with 16 main scale divisions. For this vernier calipers, the least count is **(IIT-JEE 2010)**

1. 0.02 mm
2. 0.05 mm
3. 0.1 mm
4. 0.2 mm

Question

The main scale of a Vernier callipers has n divisions/cm. n divisions of the Vernier scale coincide with $(n-1)$ divisions of main scale. The least count of the Vernier callipers is

1. $\frac{1}{(n+1)(n-1)}$ cm
2. $\frac{1}{n}$ cm
3. $\frac{1}{n^2}$ cm
4. $\frac{1}{n(n+1)}$ cm

Question

The Vernier constant of Vernier callipers is 0.1 mm and it has zero error of (–0.05) cm. While measuring diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is 5. The corrected diameter will be _____ $\times 10^{-2}$ cm.

[29 June, 2022 (Shift-II)]

Question

One main scale division of a vernier calipers is 'a' cm and nth division of the vernier scale coincide with $(n - 1)^{th}$ division of the main scale. The least count of the calipers (in mm) is **[16 March, 2021 (Shift-I)]**

1. $\frac{10na}{n-1}$
2. $\frac{10a}{n-1}$
3. $\frac{10a}{n}$
4. $\left(\frac{n-1}{10n}\right)a$

Thank

You